

Problem Description

Single-particle tracking (SPT) is widely used to study the dynamics of membrane proteins in living cells. The diffusivity of a membrane protein reflects its physical and functional state, such as monomer, dimer or oligomer, which often cause significant changes in mobility. For example, when proteins form larger complexes, their diffusion typically becomes slower. However, detecting these diffusivity changes in real time is challenging because diffusion trajectories are stochastic and noisy, making it difficult to distinguish true state transitions from random fluctuations. In particular, identifying hidden transitions between multiple hidden diffusive states (e.g., monomeric, dimeric and oligomeric states) from the single molecule trajectory data requires robust statistical inference beyond conventional mean-squared displacement (MSD) analysis. This project focuses on probabilistic approaches, such as Hidden Markov Model (HMM) and diffusion mixture model (DMM), to infer time-dependent diffusivity changes from SPT trajectories. By connecting hidden diffusion states to molecular assembly processes, these methods can provide insight into membrane protein dimerization/oligomerization kinetics at the single-molecule level.

Main Algorithm

The motion of membrane proteins can change over time between different diffusive behaviors, such as slow, intermediate, or fast diffusion. However, these diffusion states cannot be observed directly in experiments. Instead, only the jumping distances of the protein between consecutive time points can be measured. The observed sequence of jumping distances is denoted as $\mathbf{r} = [r_1, r_2, \dots, r_N]$ where measurements are taken at equally spaced time intervals (Δt). To infer the underlying diffusion behavior from these observations, we used an HMM. In this framework, the membrane protein is assumed to occupy one of K hidden diffusive states at each time point. The hidden state sequence is represented as $\mathbf{s} = [s_1, s_2, \dots, s_N]$. Each hidden state corresponds to a distinct diffusion coefficient, representing a different mobility regime of the protein. In this project, the number of hidden diffusive states, K , is assumed to be known in advance.

The goal of the HMM is to estimate the set of model parameters that maximizes the probability of observing both the hidden state sequence and the measured jumping distances: $(\hat{\boldsymbol{\pi}}, \hat{\mathbf{A}}, \hat{\mathbf{D}}) = \text{argmax}(p(\mathbf{r}, \mathbf{s} | \boldsymbol{\pi}, \mathbf{A}, \mathbf{D}))$, where $\boldsymbol{\pi}$ represents the initial probability of starting in each diffusive state, $\mathbf{A}$ is the state transition matrix describing the probability of switching between states, and $\mathbf{D}$ denotes the diffusion coefficients associated with each state. The joint probability can be decomposed into two parts: $p(\mathbf{r}, \mathbf{s} | \boldsymbol{\pi}, \mathbf{A}, \mathbf{D}) = p(\mathbf{s} | \boldsymbol{\pi}, \mathbf{A}) \cdot p(\mathbf{r} | \mathbf{s}, \mathbf{D})$, where $p(\mathbf{s} | \boldsymbol{\pi}, \mathbf{A})$ describes the probability of the hidden state sequence, and $p(\mathbf{r} | \mathbf{s}, \mathbf{D})$ describes the probability of observing the measured jumping distances given the hidden diffusion states.

- (1) The hidden state probability is determined by the initial state distribution and the transition probabilities between states: $p(\mathbf{s} | \boldsymbol{\pi}, \mathbf{A}) = \prod_{k=1}^K \pi_k^{\delta_{k,s_1}} \prod_{n=2}^N \prod_{i,j=1}^K A_{ij}^{\delta_{i,s_{n-1}} \delta_{j,s_n}}$. This formulation assumes the Markov property, meaning that the current diffusion state depends only on the previous state.
- (2) For the observed motion, the jumping distance at each time point is assumed to follow two-dimensional Brownian diffusion: $p(\mathbf{r} | \mathbf{s}, \mathbf{D}) = \prod_{n=1}^N \frac{2r_n}{4D_{s_n}\Delta t} e^{-\left(\frac{r_n^2}{4D_{s_n}\Delta t}\right)}$. This equation reflects that larger diffusion coefficients increase the probability of observing larger displacements.

A. Initialization of HMM Parameters

The model parameters of the HMM can be estimated using the Baum-Welch algorithm, a specialized expectation-maximization (EM) algorithm that requires initial parameter estimates ($\boldsymbol{\pi}_{init}$, $\mathbf{A}_{init}$, $\mathbf{D}_{init}$) before optimization.

A-1. Initialization ($\boldsymbol{\pi}_{init}$ and $\mathbf{D}_{init}$)

The initial starting probability of the HMM ($\boldsymbol{\pi}_{init}$) and initial diffusion coefficients for each diffusive state ($\mathbf{D}_{init}$) also can be determined through optimization of the DMM using the EM algorithm. If we know which state generated each jump and ignore the transition, the complete-data likelihood would be expressed as a following equation: $p(\mathbf{r}, \mathbf{s} | \boldsymbol{\pi}, \mathbf{D}) = \prod_{n=1}^N \prod_{k=1}^K [\pi_k p(r_n | D_k)]^{\delta_{k,s_n}}$. Taking the log: $\log p(\mathbf{r}, \mathbf{s} | \boldsymbol{\pi}, \mathbf{D}) = \sum_{n=1}^N \sum_{k=1}^K \delta_{k,s_n} [\log \pi_k + \log p(r_n | D_k)]$. In EM, we do not know δ_{k,s_n} , so we replace it with its posterior expectation, called the responsibility: $\gamma_{nk} = p(s_n = k | r_n, \boldsymbol{\pi}_{old}, \mathbf{D}_{old})$. Thus, the EM objective function can be expressed as follow: $Q(\boldsymbol{\pi}, \mathbf{D}) = \mathbb{E}[\log p(\mathbf{r}, \mathbf{s} | \boldsymbol{\pi}, \mathbf{D})] = \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} [\log \pi_k + \log p(r_n | D_k)]$.

- (1) In E-step, we calculate the responsibilities: $\gamma_{nk} = \frac{\pi_{k,old} p(r_n | D_{k,old})}{\sum_{j=1}^K \pi_{j,old} p(r_n | D_{j,old})}$.
- (2) In M-step, we maximize $Q(\boldsymbol{\pi}, \mathbf{D})$ with respect to $\boldsymbol{\pi}$ and $\mathbf{D}$. Here, the detailed derivation is omitted for brevity.
 - a. $\pi_{k,new} = \frac{1}{N} \sum_{n=1}^N \gamma_{nk}$.
 - b. $D_{k,new} = \frac{\sum_{n=1}^N \gamma_{nk} r_n^2}{4\Delta t \sum_{n=1}^N \gamma_{nk}}$.
- (3) The E-step and M-step are repeated until the likelihood no longer improves substantially.

A-2. Initialization ($\mathbf{A}_{init}$)

Assuming that each diffusive state is relatively stable, we initialized the transition matrix $\mathbf{A}_{init}$ by setting its diagonal elements to 0.95. The off-diagonal elements were uniformly assigned as $(1-0.95)/(K-1)$, ensuring that each row of $\mathbf{A}_{init}$ sums to one.

B. Optimization of HMM Parameters via the Baum-Welch Algorithm

Starting with initial HMM parameters $\boldsymbol{\theta}_{init} = (\boldsymbol{\pi}_{init}, \mathbf{A}_{init}, \mathbf{D}_{init})$, the likelihood of the measured single particle trajectories from each cell was maximized using Baum-Welch algorithm to obtain the model parameters $\boldsymbol{\theta}_{opt} = (\boldsymbol{\pi}_{opt}, \mathbf{A}_{opt}, \mathbf{D}_{opt})$.

B-1. Forward-Backward Algorithm

The forward-backward algorithm efficiently computes the posterior probabilities of hidden states by combining information from both directions of the trajectory. The forward variable is defined as follow: $\alpha_n(k) = p(r_1, r_2, \dots, r_n, s_n = k) = p(r_n | D_k) \sum_{i=1}^K \alpha_{n-1}(i) A_{ik}$. This is the probability of observing the trajectory up to time point n and ending in state k . The backward variable is defined as follow: $\beta_n(k) = p(r_{n+1}, r_{n+2}, \dots, r_N | s_n = k) = \sum_{j=1}^K A_{kj} p(r_{n+1} | D_j) \beta_{n+1}(j)$. This is the probability of observing the remaining trajectory from time $n+1$ to N , assuming that the system is currently in state k at time n . Both α and β can be computed recursively.

B-2. Main EM Algorithm

By multiplying the forward and backward variables, we can evaluate how likely it is that the system was in state k at time n , while considering the full observed sequence: $\gamma_n(k) = p(s_n = k | \mathbf{r}, \boldsymbol{\pi}, \mathbf{A}, \mathbf{D}) = \frac{\alpha_n(k) \beta_n(k)}{\sum_{l=1}^K \alpha_n(l) \beta_n(l)}$. Similarly, the probability of a transition from state i to j is computed as follow: $\xi_n(i, j) = p(s_n = i, s_{n+1} = j | \mathbf{r}, \boldsymbol{\pi}, \mathbf{A}, \mathbf{D}) = \frac{\alpha_n(i) A_{ij} p(r_{n+1} | D_j) \beta_{n+1}(j)}{\sum_{i=1}^K \sum_{j=1}^K \alpha_n(i) A_{ij} p(r_{n+1} | D_j) \beta_{n+1}(j)}$. Thus, the forward-backward algorithm allows the HMM to estimate hidden-state probabilities by considering not only the displacement at a single time point, but also the temporal context before and after that point, given the prior knowledge. The EM objective function can be expressed as follow: $Q(\boldsymbol{\pi}, \mathbf{A}, \mathbf{D}) = \mathbb{E}[\log p(\mathbf{r}, \mathbf{s} | \boldsymbol{\pi}, \mathbf{A}, \mathbf{D})] = \mathbb{E}[\log \pi_{s_1} + \sum_{n=2}^N \log A_{s_{n-1} s_n} + \sum_{n=1}^N \log p(r_n | D_{s_n})] = \sum_{k=1}^K \gamma_1(k) \log \pi_k + \sum_{n=1}^{N-1} \sum_{i=1}^K \sum_{j=1}^K \xi_n(i, j) \log A_{ij} + \sum_{n=1}^N \sum_{k=1}^K \gamma_n(k) \log p(r_n | D_k)$.

- (1) In E-step, we calculate the responsibilities: $\gamma_n(k) = \frac{\alpha_n(k) \beta_n(k)}{\sum_{l=1}^K \alpha_n(l) \beta_n(l)}$ and $\xi_n(i, j) = \frac{\alpha_n(i) A_{ij,old} p(r_{n+1} | D_{j,old}) \beta_{n+1}(j)}{\sum_{i=1}^K \sum_{j=1}^K \alpha_n(i) A_{ij,old} p(r_{n+1} | D_{j,old}) \beta_{n+1}(j)}$ with old $\alpha_n(k)$ and $\beta_n(k)$.
- (2) In M-step, we maximize $Q(\boldsymbol{\pi}, \mathbf{A}, \mathbf{D})$ with respect to $\boldsymbol{\pi}$, $\mathbf{A}$ and $\mathbf{D}$. Here, the detailed derivation is omitted for brevity.
 - a. $\pi_{k,new} = \gamma_1(k)$.
 - b. $A_{ij,new} = \frac{\sum_{n=1}^{N-1} \xi_n(i, j)}{\sum_{n=1}^{N-1} \gamma_n(i)}$.
 - c. $D_{k,new} = \frac{\sum_{n=1}^N \gamma_n(k) r_n^2}{4\Delta t \sum_{n=1}^N \gamma_n(k)}$.
- (3) The E-step and M-step are repeated until the likelihood no longer improves substantially.

C. Viterbi Algorithm (Beyond the Scope of This Project)

After optimizing the HMM parameters using the Baum-Welch algorithm, the most likely hidden diffusive state sequence can be inferred using the Viterbi algorithm. The Viterbi algorithm identifies the single state path that maximizes the joint probability of the observed jumping-distance sequence and the hidden state sequence, given the optimized model parameters. In this study, the main focus was on estimating the HMM parameters, including the diffusion coefficients and state transition probabilities. Therefore, the detailed implementation and derivation of the Viterbi algorithm are beyond the scope of this project. Briefly, Viterbi decoding provides a deterministic assignment of each observed displacement to the most likely hidden diffusive state based on the learned HMM.

Description of Dataset

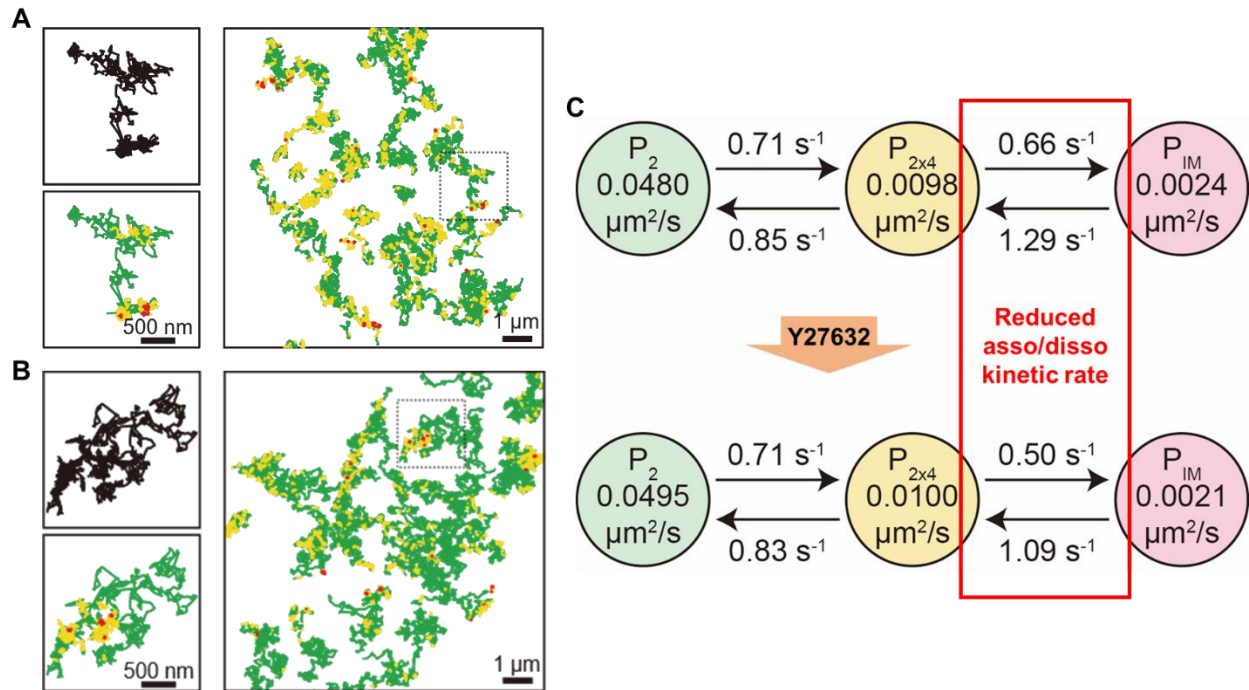
Single-molecule tracking data of the membrane protein PLVAP using optical microscopy were used in this study. Whole dataset was obtained from the previous lab. The trajectories were extracted from time-lapse images acquired at discrete time points separated by a time interval of $\Delta t = 20$ (ms).

To estimate reliable HMM parameters while assessing the degree of cell-to-cell variation, single-particle trajectories from individual cells were pooled and fit to the same HMM framework. A total of 10 cells were used for analysis, including 5 ‘control’ cells and 5 ‘Y27632-treated’ cells. On average, each cell contained approximately 54 trajectories, with an average trajectory length of approximately 450 frames ($N \approx 5 \times 54 \times 450 \approx 120,000$).

Based on prior knowledge of PLVAP oligomeric states, the number of hidden diffusive states was fixed to $K = 3$. These three states were assumed to represent distinct mobility regimes of PLVAP, such as, fast (dimer, P_2), intermediate (dimer-of-tetramer, $P_{2 \times 4}$), and slow (immobile complex, P_{IM}) diffusion states.

All computations were performed using custom codes with MATLAB 2020a software.

Results and Discussion



Representative trajectories of PLVAP under control and Y27632-treated conditions are shown as black lines in panels A and B, respectively. The Baum-Welch algorithm estimated the diffusion coefficient of each PLVAP diffusive state, as well as the transition frequency between states, as summarized in panel C.

Independent HMM analyses of the control and Y27632-treated conditions yielded similar diffusion coefficients for all three PLVAP states, suggesting that the algorithm provides robust parameter estimation and that the dataset is sufficiently large to extract information on assembly dynamics. Notably, transitions between P_2 and P_{IM} were negligible, indicating that the dynamic equilibrium of PLVAP assembly proceeds in a stepwise manner rather than through direct conversion between these two states. Upon Y27632 treatment, the transition rate between $P_{2 \times 4}$ and P_{IM} was significantly reduced.

The Viterbi algorithm was used to annotate the most probable diffusive state for each jump in the trajectories. In panels A and B, green represents the fastest diffusive state, P_2 ; yellow represents the intermediate diffusive state, $P_{2 \times 4}$; and red represents the slowest diffusive state, P_{IM} . Consistent with the diffusion coefficients inferred by the Baum-Welch algorithm, the large jumping steps in the green trajectories reflect the fast diffusive motion of P_2 , whereas the localized, stationary pattern of the red trajectories reflects the immobile behavior of P_{IM} .

This analysis was based on a custom MATLAB implementation that had previously been developed in the laboratory for HMM-based analysis of single-particle tracking data. Although the computational framework was not built entirely from scratch in this project, revisiting the existing code provided an opportunity to examine how the Baum-Welch and Viterbi algorithms were implemented in practice and to connect each computational step to its underlying mathematical formulation. Therefore, the main contribution of this project was not simply the application of an existing HMM tool, but the mathematical interpretation and reconstruction of the algorithmic logic behind the analysis.

Reference(s)

1. Yu Matsuda et al. “Estimation of diffusive states from single-particle trajectory in heterogeneous medium using machine-learning methods” Physical Chemistry Chemical Physics 2018